

Vinay Mattapalli

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SUMMARY

Innovative and results-driven **AI/ML Engineer** with over **5 years of experience** delivering scalable machine learning and GenAI solutions across healthcare and enterprise domains. Skilled in **LLM application development, MLOps, and cloud-native deployments** using Azure, AWS, and GCP. Proven track record of building **production-grade ML pipelines**, integrating **vector search, RAG architectures, and observability**, and collaborating with cross-functional teams to deliver measurable business outcomes.

SKILLS

- **Languages:** Python (Pandas, NumPy, Matplotlib, LangChain), SQL, R, Scala
- **Machine Learning:** PyTorch, TensorFlow, HuggingFace Transformers, NLP, RAG, LLM Fine-tuning, XGBoost, Time Series Forecasting, Feature Engineering, A/B Testing
- **MLOps & Deployment:** MLflow, Airflow, Docker, Kubernetes, Helm, CI/CD (Azure DevOps, GitHub Actions), FastAPI, Terraform, Weights & Biases (W&B)
- **GenAI & Vector DB:** OpenAI APIs, LangChain, Pinecone, FAISS, Azure OpenAI, Guardrails, LlamaIndex, Weaviate, Vector Search, Seldon Core
- **Cloud & Security:** Azure (AKS, ADF, AML, Synapse), AWS (S3, SageMaker, Bedrock, Lambda, CloudFormation), GCP (BigQuery, Vertex AI Pipelines, Dataflow), IAM, RBAC, Secrets Manager
- **Monitoring & Observability:** Prometheus, Grafana, EvidentlyAI
- **Data & Visualization:** Azure Data Factory, Snowflake, Kafka, Tableau, Power BI, Data Studio

EXPERIENCE

Humana

Aug 2024 – Present

Software Engineer

- Designed and deployed **LLM-powered RAG solutions** using **LangChain, Pinecone, and OpenAI APIs**, enabling real-time document retrieval and summarization that improved internal claims processing time by **30%**.
- Built and maintained **end-to-end ML pipelines** on **Azure Kubernetes Service (AKS)** and **Azure Machine Learning**, containerizing models with Docker and Helm, integrating with **CI/CD pipelines** in **Azure DevOps**, and reducing deployment cycles from weeks to days.
- Implemented **MLflow** for experiment tracking and automated retraining workflows orchestrated with **Apache Airflow**, ensuring reproducibility and governance across dev and prod environments.
- Enhanced **model observability** by integrating **Prometheus, Grafana, and EvidentlyAI**, enabling real-time performance monitoring and early drift detection in production.
- Partnered with data engineering teams to optimize ETL workflows via **Azure Data Factory** and SQL, improving data freshness and reliability for downstream ML applications.
- Developed and deployed **NLP models** for sentiment analysis and entity recognition on patient feedback datasets, supporting regulatory and customer experience reporting.
- Collaborated with cross-functional teams in a **HIPAA-regulated** environment to operationalize AI solutions with robust monitoring, security, and governance.

Bupa

Feb 2020 – Dec 2021

Software Engineer

- Engineered **ML pipelines** on **AWS** using **SageMaker, S3, and Lambda** for automated training and real-time inference, improving data processing efficiency by **50%**.
- Developed and optimized forecasting and classification models using **PyTorch, XGBoost, and HuggingFace Transformers**, achieving up to **10% improvement** in predictive performance.
- Integrated **Weights & Biases (W&B)** to track experiments and improve model reproducibility and visibility across teams.
- Containerized **ML workloads** with Docker and deployed them on **AWS ECS**, using Jenkins CI/CD pipelines for consistent deployments.
- Leveraged **RAPIDS/CUDA** to accelerate data processing tasks, reducing pipeline latency by **10%** for large datasets.
- Built real-time feature pipelines using MongoDB and Kafka to enable low-latency model updates.
- Ensured model deployment practices aligned with internal **cloud security and compliance standards**.

- Built and deployed supervised learning models using **TensorFlow** and **PyTorch**, improving model accuracy by **15%** through advanced feature engineering and hyperparameter tuning.
- Designed **automated ML training pipelines** on **GCP (Cloud ML Engine + Compute Engine)** with CI/CD via Cloud Build and Git, ensuring robust version control and faster release cycles.
- **Reduced model deployment cycles by 40%** through infrastructure automation and standardized pipeline templates, improving collaboration between ML and DevOps teams.
- Developed **interactive dashboards** in Tableau and SQL to support operational reporting, reducing manual data preparation by **30%**.
- Implemented large-scale data processing workflows using **Dataproc (Hadoop/Spark)** to optimize feature engineering for downstream ML tasks.
- Utilized **BigQuery** and Data Studio to automate data extraction and build dynamic reports for internal stakeholders.

EDUCATION/ CERTIFICATIONS

- Master of Science in Information Science, University of North Texas, Denton, TX
- Master of Business Administration, University of Sunderland, Sunderland, United Kingdom
- IBM AI Developer Certification
- Building Generative AI-Powered Applications with Python

PROJECTS

- **RAG Chatbot Platform:** Built a Retrieval-Augmented Generation chatbot using LangChain, LlamaIndex, Pinecone, and OpenAI APIs, deployed on AKS with FastAPI and monitored via Prometheus and EvidentlyAI.
- **Model Serving with Seldon:** Deployed PyTorch models using Seldon Core on Kubernetes with Helm, enabling scalable inference and canary rollouts for A/B testing.
- **Vector Search Engine:** Implemented a hybrid semantic search engine using Weaviate and LangChain to support enterprise document retrieval use cases.